

Life Cycle Assessment of Small-Scale Tofu Production in Semarang City: Hotspot Analysis Using CML-IA and AWARE Methods

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Abstract

The tofu industry is one of the most important food-processing sectors in Indonesia, given its high consumption rate and notable nutritional value. However, the environmental impacts of each production stage have not been comprehensively identified. Life Cycle Assessment (LCA) with a gate-to-gate system boundary was applied to evaluate the environmental impacts of tofu production at a small and medium-sized enterprise (SME) in Semarang City. The LCA was conducted using SimaPro 10.3 software with the CML-IA Baseline impact assessment method, covering five impact categories: Global Warming Potential (GWP), Ozone Layer Depletion (ODP), Fresh Water Aquatic Ecotoxicity (FAET), Acidification, and Eutrophication. Water use was assessed using the AWARE method. Results demonstrate that the washing and boiling stages are the primary contributors to environmental impacts. Normalization results indicate that Fresh Water Aquatic Ecotoxicity is the most dominant impact category (1.33×10^3 kg 1,4-DB eq), followed by Global Warming Potential (5.28×10^3 kg CO₂ eq). The main environmental hotspots are organic wastewater from washing, soaking, and coagulation stages, and firewood combustion during boiling. Recommended improvement strategies include wastewater treatment systems, water recycling, and fuel substitution.

Keywords: Life Cycle Assessment; tofu production; CML-IA Baseline; AWARE; environmental hotspot; Semarang

1. Introduction

Industrial activity is one of the primary drivers of global economic growth [1]. The tofu industry represents a significant food-processing sector in Indonesia owing to its high consumption rate and notable nutritional value. According to Statistics Indonesia (BPS), average per capita tofu consumption reached approximately 0.163 kg per week in 2024, indicating that tofu remains a staple food in many Indonesian households [2]. This high demand has driven increased production activity, resulting in greater resource and energy consumption, as well as waste and emissions that may cause environmental pollution if not properly managed [3].

Such impacts are particularly evident in river water quality, as the discharge of industrial wastewater can cause Biochemical Oxygen Demand (BOD) and Chemical Oxygen Demand (COD) levels to exceed regulatory standards. In the tofu industry in Mojokerto, liquid waste was reported to contain BOD exceeding 2,000 mg/L and COD exceeding 5,000 mg/L — far above the standard thresholds of approximately 50–100 mg/L [4]. Furthermore, the environmental impacts associated with each tofu production stage have not yet been comprehensively identified. A

systematic approach such as Life Cycle Assessment (LCA) is therefore required to evaluate the full environmental burden of the tofu production process [5].

Life Cycle Assessment (LCA) is a standardized method used to systematically evaluate the environmental impacts of a product or process based on all input and output flows throughout its life cycle [6]. This method identifies production stages that constitute environmental hotspots, thereby serving as the scientific basis for developing targeted environmental improvement strategies [7]. Previous LCA studies of the tofu industry predominantly focused on climate change categories and thus failed to comprehensively evaluate other environmental concerns such as water pollution, water scarcity, and solid waste generation across all production stages [8, 9].

Therefore, this study aims to analyze the environmental impacts of tofu production in Semarang City using an LCA approach with a gate-to-gate system boundary — encompassing all stages from washing to packaging — in order to identify environmental hotspots and formulate more sustainable improvement strategies.

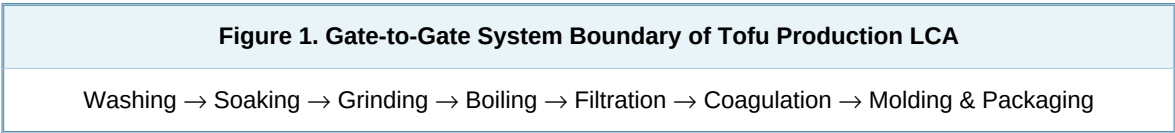
2. Methods

This research utilized a quantitative LCA methodology to assess the environmental impacts of tofu production at an SME in Semarang City. The evaluation was performed in compliance with **SNI ISO 14040:2016** and **SNI ISO 14044:2016**, comprising four sequential phases: (1) goal and scope definition, (2) life cycle inventory analysis, (3) life cycle impact assessment, and (4) interpretation. Primary data were obtained through direct field measurements, structured interviews, and on-site documentation during a complete production batch. Secondary data were sourced from peer-reviewed literature and the Ecoinvent v3 database within SimaPro 10.3 [6].

2.1 Goal and Scope Definition

This study assesses the environmental impacts of tofu production at the facility level, focusing on water use, energy consumption, and waste generation at each production stage. The **functional unit** is defined as **1 kg of tofu produced**, consistent with comparable LCA studies of tofu and soy-based food processing. Inventory data were collected on a per-batch basis: one full production day with 1,400 kg of soybeans as raw material input, yielding 3,681 kg of tofu.

The **system boundary** adopts a gate-to-gate approach encompassing all unit processes from initial soybean washing to final molding and packaging (Figure 1). Upstream operations (soybean farming, transportation) and downstream activities (distribution, consumption, end-of-life) were excluded, consistent with prior gate-to-gate LCA investigations of similar Indonesian tofu SMEs [8, 9].



2.2 Life Cycle Inventory (LCI) Analysis

The inventory analysis phase involves identification and quantitative collection of all input and output flows for each unit process [11]. Data include soybean inputs, water and energy consumption, and waste outputs (liquid waste, solid waste, and atmospheric emissions). The full

inventory is presented in Table 1.

Table 1. Life Cycle Inventory Data of Tofu Production (per batch / production day).

Stage	Input	Qty	Unit	Output	Qty	Unit
Washing	Soybeans	1,400	kg	Washed soybeans	1,540	kg
	Water	2,800	L	Wastewater	2,660	L
	Electricity	0.60	kWh	CO ₂ emissions	0.51	kg
Soaking	Soybeans	1,540	kg	Hydrated soybeans	3,640	kg
	Water	2,700	L	Wastewater	600	L
	Electricity	0.45	kWh	CO ₂ emissions	0.38	kg
Grinding	Hyd. soybeans	3,640	kg	Soybean slurry	7,840	kg
	Water	4,200	L	Wastewater	0	L
	Electricity	17.00	kWh	CO ₂ emissions	14.45	kg
Boiling	Soybean slurry	7,840	kg	Boiled slurry	15,652	kg
	Water	8,400	L	Steam (evap.)	588	L
	Firewood	800	kg	CO ₂ emissions	1,397	kg
				CH ₄ emissions	0.050	kg
				N ₂ O emissions	0.374	kg
Filtration	Boiled slurry	15,652	kg	Soy milk	14,052	kg
				Okara (residue)	1,600	kg
Coagulation	Soy milk	14,052	kg	Tofu curd	3,681	kg
	Coagulant water	421	L	Whey	10,434	L
Molding & Pkg	Tofu curd	3,681	kg	Final product	3,681	kg
	Water	100	L	Wastewater	100	L

2.3 Life Cycle Impact Assessment (LCIA)

The LCIA phase evaluates and quantifies potential environmental impacts generated by the product system [6]. The **CML-IA Baseline** method was applied via SimaPro 10.3, covering: Global Warming Potential (GWP₁₀₀, kg CO₂ eq), Ozone Layer Depletion (ODP, kg CFC-11 eq), Fresh Water Aquatic Ecotoxicity (FAET, kg 1,4-DB eq), Acidification (AP, kg SO₂ eq), and Eutrophication (EP, kg PO₄³⁻ eq). Water use and potential water scarcity were additionally assessed using the **AWARE (Available Water REmaining)** method.

2.4 Interpretation

The interpretation phase contextualizes LCIA results to identify hotspot categories and production stages contributing the greatest environmental burdens [12]. **Normalization** was applied using CML-IA Baseline reference values to place all impact categories on a common dimensionless scale, enabling direct comparison of relative environmental significance.

3. Results and Discussion

3.1 Environmental Impact of Tofu Production

The LCIA reveals measurable environmental impacts across all five assessed categories. Results are summarized in Table 2. Note that the filtration stage does not appear in the impact results because no direct energy inputs or emissions were entered in the inventory model for this stage; SimaPro therefore generates no characterization values for filtration.

Table 2. Environmental Impact of Tofu Production per Batch by Production Stage.

Impact Category	Unit	Total	Washing	Soaking	Grinding	Boiling	Coagulation	Molding
GWP	kg CO ₂ eq	5.28×10 ³	3.56×10 ³	0.945	35.8	1.63×10 ³	55.8	5.9
ODP	kg CFC-11 eq	6.11×10 ³	5.93×10 ³	2.1×10 ⁻³	7.93×10 ⁻³	1.07×10 ⁻³	5.02×10 ⁻³	1.29×10 ⁻³
FAET	kg 1,4-DB eq	1.33×10 ³	1.22×10 ³	0.651	24.6	70.6	14.1	2.75
Acidification	kg SO ₂ eq	4.59	3.53	0.00245	0.0925	0.723	0.226	0.0176
Eutrophication	kg PO ₄ ³⁻ eq	9.10	8.30	0.0034	0.128	0.409	0.250	0.00838

3.1.1 Global Warming Potential (GWP)

Total GWP amounts to 5.28×10³ kg CO₂ eq. The washing stage is the largest contributor at 3.56×10³ kg CO₂ eq (≈67.4%), followed by the boiling stage at 1.63×10³ kg CO₂ eq (≈30.9%). The dominance of washing is attributed to large water volumes consumed, which indirectly require energy for water supply and distribution, generating greenhouse gas emissions. Organic-laden washing wastewater also has the potential to produce methane (CH₄) if discharged untreated. The boiling stage contributes significantly through firewood combustion, releasing CO₂, CH₄, and N₂O — all potent greenhouse gases.

3.1.2 Ozone Layer Depletion (ODP)

Total ODP registers 6.11×10³ kg CFC-11 eq, with washing dominating at 5.93×10³ kg CFC-11 eq (≈97.1%). This impact is not attributed to water itself but to the energy-intensive infrastructure supporting water supply and distribution, which generates indirect emissions linked to ozone-depleting substances within the LCA system boundary.

3.1.3 Fresh Water Aquatic Ecotoxicity (FAET)

Total FAET reaches 1.33×10³ kg 1,4-DB eq. The washing stage is the primary contributor at 1.22×10³ kg 1,4-DB eq (≈91.7%). The large volumes of organic-laden wastewater from soybean washing — containing residual organic matter, dissolved solids, and impurities — constitute a significant toxicity threat to freshwater organisms if discharged without treatment. The boiling stage contributes 70.6 kg 1,4-DB eq (≈5.3%) through atmospheric deposition of combustion byproducts (NO_x, SO_x) into aquatic environments.

3.1.4 Acidification

Total Acidification impact is 4.59 kg SO₂ eq. The washing stage accounts for 3.53 kg SO₂ eq (≈76.9%), linked to SO₂ and NO_x emissions from energy generation supporting water provision. The boiling stage contributes 0.723 kg SO₂ eq (≈15.8%) through firewood combustion, producing gases that may form acid rain and degrade soil and water quality.

3.1.5 Eutrophication

Total Eutrophication impact is 9.10 kg PO₄³⁻ eq, with washing dominant at 8.30 kg PO₄³⁻ eq (~91.2%). This is driven by nutrient-rich washing wastewater containing nitrogen and phosphorus compounds from soybean residues. Discharge of these nutrients can trigger excessive algal proliferation, depleting dissolved oxygen and disrupting aquatic ecosystems. The boiling stage contributes 0.409 kg PO₄³⁻ eq (~4.5%) through atmospheric NO_x deposition.

3.2 Water Use Assessment (AWARE Method)

Water use was assessed using the AWARE method to determine potential water scarcity across all production stages. Total water scarcity impact is **515 m³ world eq** for the entire production system (Table 3).

Table 3. Water Use Assessment Results per Production Stage (AWARE Method).

Production Stage	Water Scarcity (m ³ world eq)	Share (%)
Boiling	294	57.09
Grinding	101	19.61
Soaking	39.3	7.63
Coagulation	38.2	7.42
Washing	35.8	6.95
Molding & Packaging	7.63	1.48
Total	515	100

The boiling stage is the dominant contributor (~57.1%), reflecting both direct water consumption and the substantial upstream water demand associated with firewood supply chains. The grinding stage contributes the second largest share (~19.6%), as continuous water addition is required to achieve appropriate slurry consistency. Within the AWARE framework, the environmental burden is determined not solely by volumetric water use but also by the upstream energy system and supply chain characteristics of each production stage.

3.3 Normalization

Normalization was performed using CML-IA Baseline reference values to place all impact categories on a common dimensionless scale, enabling direct comparison of relative environmental significance (Table 4).

Table 4. Normalization Results of Tofu Production Environmental Impacts (CML-IA Baseline).

Impact Category	Normalized Value	Rank
Fresh Water Aquatic Ecotoxicity (FAET)	2.57×10 ⁻¹	1
Global Warming Potential (GWP)	1.05×10 ⁻¹	2
Eutrophication	6.89×10 ⁻²	3
Acidification	1.63×10 ⁻²	4
Ozone Layer Depletion (ODP)	6.84×10 ⁻¹³	5

Normalization confirms that **Fresh Water Aquatic Ecotoxicity (FAET)** is the dominant environmental impact category, indicating that tofu production exerts the greatest pressure on

freshwater ecosystems. Although GWP frequently receives priority attention in sustainability discussions, its normalized contribution ranks second, indicating that freshwater pollution represents a more acute concern at the facility scale assessed. Eutrophication (rank 3) further reinforces the urgency of addressing nutrient-laden wastewater streams.

3.4 Hotspot Identification and Improvement Strategies

Based on the integrated LCIA and normalization results, two primary environmental hotspots are identified:

Hotspot 1: Organic Liquid Waste (Washing, Soaking, Coagulation Stages)

The organic-laden wastewater streams from washing, soaking, and coagulation (whey) are the principal drivers of FAET and Eutrophication — ranked first and third in normalized impact. These effluents are characterized by high BOD, COD, and nutrient loads [13, 14]. Recommended interventions:

- *Anaerobic treatment systems* (biodigester/IPAL biogas) to reduce BOD/COD while recovering biogas energy
- *Constructed wetlands or biofilter systems* for nutrient removal prior to discharge
- *Water recycling and recirculation* within washing and soaking stages to reduce volumetric discharge
- *Whey valorization* (e.g., as animal feed or fermentation substrate) to divert high-nutrient streams

Hotspot 2: Firewood Combustion (Boiling Stage)

The boiling stage dominates GWP, AWARE water scarcity, and is the second-largest contributor to acidification and eutrophication. Combustion of 800 kg firewood per batch releases 1,397 kg CO₂, 0.0499 kg CH₄, and 0.3744 kg N₂O. Recommended interventions:

- *Fuel substitution* to LPG, biomethane from on-site biodigesters, or high-efficiency biomass systems
- *Improved boiler or insulated boiling tank systems* to minimize heat loss and reduce fuel consumption per batch
- *Waste heat recovery* from boiling exhaust to pre-heat water in upstream stages

The LCA-based approach demonstrates that improvements targeting liquid waste management at the source yield greater environmental reduction benefits than end-of-pipe interventions [10, 15]. Prioritizing these two hotspots is expected to substantially reduce the overall environmental footprint of tofu production in Semarang City.

4. Conclusion

Based on the Life Cycle Assessment of tofu production in Semarang City using a gate-to-gate approach, the following conclusions are drawn:

- **Tofu production generates measurable environmental impacts** across all five assessed categories: GWP (5.28×10^3 kg CO₂ eq), ODP (6.11×10^{-4} kg CFC-11 eq), FAET (1.33×10^3 kg 1,4-DB eq), Acidification (4.59 kg SO₂ eq), and Eutrophication (9.10 kg PO₄³⁻ eq).

- **The washing and boiling stages are the primary contributors** to environmental impact across all categories, driven by high water consumption and firewood combustion, respectively.
- **Fresh Water Aquatic Ecotoxicity (FAET) is the dominant impact category** based on normalization results (2.57×10^3), confirming that organic liquid waste — particularly washing effluents, whey, and coagulation waste — constitutes the most critical environmental hotspot.
- **Water scarcity** (AWARE method) totals 515 m³ world eq, with the boiling stage as dominant contributor (57.1%), largely driven by the upstream water demand of firewood supply chains.
- **Improvement strategies** should prioritize: (a) implementation of wastewater treatment and recycling systems for washing, soaking, and coagulation effluents; and (b) substitution of firewood with cleaner fuel alternatives to reduce greenhouse gas emissions and water scarcity impacts.

The findings provide actionable, evidence-based guidance for improving the environmental sustainability of SME-scale tofu production and can inform environmental management policies for the tofu industry in Semarang City and Central Java.

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